

Fairness-Aware Loss Function Design for Healthcare Risk Prediction

Rishi Modi, Blessing Ogbuokiri

Responsible and Applied Machine Learning Laboratory (RAML Lab),
Department of Computer Science, Brock University
St. Catharines, ON L2S 3A1, Canada.
rm22co@brocku.ca, bogbuokiri@brocku.ca

Abstract

Fairness-aware loss functions that penalize group-level prediction disparities during training are increasingly applied to clinical risk prediction models, yet their effectiveness varies substantially across datasets and clinical contexts in ways that are not well understood prior to deployment. We propose a pre-training diagnostic framework based on two measurable dataset structural properties, Base Rate Disparity (BRD) and Positive Class Sparsity (PCS), that provide empirical indicators of whether fairness regularization may find a viable accuracy-fairness operating region, encounter structural optimization conflict, or prove unnecessary for a given dataset. Validated across five datasets spanning healthcare and non-healthcare domains, pre-training predictions derived from BRD and PCS were consistently associated with observed optimization behavior, including a viable operating region on the UCI Heart Disease dataset, sustained conflict on Adult Income, and unnecessary degradation on three datasets where baseline disparities were already limited. Building on this diagnostic, we propose an adaptive fairness-aware loss function that automatically scales regularization strength according to dataset structure, re-framing the role of adaptive fairness methods as optimization governance rather than fairness maximization, and demonstrate that this formulation reduces avoidable accuracy degradation without sacrificing achievable fairness improvements.

Keywords: algorithmic fairness, fairness-aware machine learning, clinical risk prediction, regularization, healthcare AI

1 Introduction

The deployment of machine learning models in clinical decision support has created renewed urgency around algorithmic fairness. Predictive models used for risk stratification, readmission prediction, and disease diagnosis can systematically produce disparate outcomes across demographic groups, potentially exacerbating existing health inequities rather than alleviating them (Pfohl, Foryciarz, and Shah 2021; Obermeyer et al. 2019). The integration of machine learning into clinical workflows has accelerated substantially (Rajkomar et al. 2018), amplifying the urgency of ensuring equitable model behavior across patient populations. In response, the machine learning community has developed a family of fairness-aware training methods that augment

standard prediction objectives with penalty terms designed to reduce group-level disparities during optimization (Zafar et al. 2017; Hardt, Price, and Srebro 2016). These methods are increasingly treated as standard tools for responsible clinical ML deployment. Misapplied fairness regularization in healthcare settings may degrade predictive reliability without producing meaningful equity gains, introducing deployment risk and complicating clinical governance.

However, a critical assumption underlies their practical use: that fairness regularization is broadly applicable across datasets and clinical contexts. A practitioner deploying a readmission prediction model applies a fairness penalty and tunes its strength, implicitly assuming that the optimization will find a viable accuracy-fairness trade-off. This assumption is rarely examined before training begins. Foundational work has shown that common fairness criteria are mathematically incompatible when group outcome rates differ (Chouldechova 2017; Kleinberg, Mullainathan, and Raghavan 2017), a constraint that directly motivates examining the structural conditions under which fairness optimization is viable. The empirical evidence suggests it should be. Berk et al. (2017) studied fairness regularization across six datasets and found that accuracy-fairness trade-offs vary substantially, concluding that these trade-offs lack prescriptive universals. Pfohl, Foryciarz, and Shah (2021) extended this observation to clinical risk prediction specifically, finding heterogeneous outcomes across hospital databases and clinical conditions. Neither study identified a structural explanation for when and why these outcomes differ.

This paper addresses that gap directly. We argue that fairness-aware optimization viability is not a property of the regularization method but of the dataset it encounters, specifically, of two measurable structural properties that can be computed before any model training begins. The first, *Base Rate Disparity* (BRD), measures the difference in positive-class rates between demographic groups and predicts the severity of gradient conflict under Demographic Parity regularization. The second, *Positive Class Sparsity* (PCS), measures the proportion of minority-group positive-class samples and predicts the effective gradient signal available to Equal Opportunity regularization. Validated across five datasets spanning healthcare and non-healthcare domains, pre-training predictions from BRD and PCS consistently aligned with observed outcomes, including a vi-

able operating region on Heart Disease, sustained conflict on Adult Income, and unnecessary degradation on three datasets where baseline disparities were already limited.

Building on this diagnostic, we propose an adaptive fairness-aware loss function that automatically scales regularization strength according to dataset structural conditions, preventing the application of fairness pressure where it cannot be productively absorbed. We show that this adaptive formulation reduces unnecessary accuracy degradation across datasets without sacrificing achievable fairness improvements.

The contributions of this paper are threefold: (1) BRD and PCS as pre-training diagnostic quantities associated with fairness regularization behavior across datasets; (2) a three-regime taxonomy of viable, conflict, and low-necessity optimization outcomes; and (3) an adaptive loss function that scales fairness pressure according to dataset structure, framing adaptive fairness as optimization governance rather than fairness maximization.

2 Related Work

The formal study of algorithmic fairness traces back to foundational definitions of individual fairness and awareness-based equitable treatment (Dwork et al. 2012), which established the theoretical grounding upon which subsequent fairness-aware optimization approaches build (Barocas, Hardt, and Narayanan 2019). We organize the related work along three dimensions relevant to this study: fairness regularization frameworks, empirical heterogeneity in fairness trade-offs, and theoretical context for regularizer behavior.

Fairness Regularization Frameworks. The fairness-aware optimization paradigm underlying this work was established by Zafar et al. (2017), who introduced constraint-based mechanisms for enforcing Demographic Parity in classification through differentiable penalty terms added to the training objective. Hardt, Price, and Srebro (2016) formalized Equal Opportunity as a fairness criterion requiring equalization of true positive rates across demographic groups, providing the theoretical basis for the EO penalty formulation used in this work.

Subsequent work extended fairness-aware learning to prejudice-aware and reductions-based optimization frameworks (Kamishima et al. 2012; Agarwal et al. 2018).

Empirical Heterogeneity in Fairness Trade-offs. A recurring observation in the fairness literature is that accuracy–fairness trade-offs vary substantially across datasets and application domains, yet the structural causes of this variation remain incompletely understood (Mehrabi et al. 2021; Corbett-Davies and Goel 2018). Berk et al. (2017) conducted an extensive comparative study of fairness regularization across six datasets, introducing the *Price of Fairness* as a measure of the accuracy cost of imposing fairness constraints.

Their results demonstrated that this cost varies considerably across datasets, leading them to conclude that fairness trade-offs lack prescriptive universals. While this observation is central to the motivation for our work, Berk et al.

(2017) do not identify the dataset-level structural properties responsible for the observed variation.

Pfohl, Foryciarz, and Shah (2021) extended this empirical characterization to clinical risk prediction specifically, studying the impact of group fairness constraints across multiple observational healthcare databases, clinical outcomes, and sensitive attributes. Their results confirmed that fairness-aware optimization produces heterogeneous outcomes across clinical settings, in some cases substantially degrading predictive performance without meaningful fairness improvement, and in others achieving fairness gains at acceptable cost, but again without a structural explanation for when and why these outcomes differ.

The diagnostic framework proposed in this work directly addresses the gap left by both studies: Base Rate Disparity and Positive Class Sparsity provide pre-training structural explanations for the heterogeneity that Berk et al. (2017) and Pfohl, Foryciarz, and Shah (2021) observed empirically.

Prior work has explored **adaptive fairness optimization** through dynamic regularization adjustment, fairness-aware hyperparameter scheduling, and multi-objective optimization (Cotter, Jiang, and Sridharan 2019; Martinez, Bertran, and Sapiro 2020; Sener and Koltun 2018). Unlike these approaches, which adapt during training, our framework performs pre-training structural diagnosis using dataset statistics alone.

Finocchiario (2023) theoretically characterized conditions under which fairness regularizers alter the minimizer of the training objective. Our work complements this perspective empirically by identifying dataset-level structural properties associated with when regularization exerts meaningful optimization influence. We do not claim BRD and PCS directly instantiate these theoretical conditions, but the observed behavior is consistent with fairness regularizers interacting with dataset structure.

3 Methodology

3.1 Fairness-Aware Learning Framework

We consider a standard binary classification setting in which a model learns to predict a binary outcome $y \in \{0, 1\}$ from a feature vector $\mathbf{x} \in \mathbb{R}^d$. Each instance is additionally associated with a sensitive attribute $A \in \{0, 1\}$ representing the two demographic groups under consideration. The model produces a predicted probability

$$\hat{y} = \sigma(\mathbf{w}^T \mathbf{x} + b) \quad (1)$$

where $\sigma(\cdot)$ is the sigmoid function and $\theta = (\mathbf{w}, b)$ denotes the model parameters optimized via gradient descent. The standard predictive objective minimizes binary cross-entropy loss over N training samples:

$$L_{\text{pred}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2)$$

Optimizing L_{pred} alone does not account for demographic fairness and may produce systematic disparities across

groups defined by the sensitive attribute. Note that the sensitive attribute A is used only to evaluate and regularize the model; it is excluded from the feature vector $\mathbf{x}$ throughout this work, so any disparity that emerges is learned indirectly through correlated features rather than direct conditioning on group membership.

To incorporate fairness during training, we introduce regularization terms that penalize group-level prediction disparities. Let $G_0 = \{i : A_i = 0\}$ and $G_1 = \{i : A_i = 1\}$ denote the two demographic groups, and let $G_0^+ = \{i : y_i = 1, A_i = 0\}$ and $G_1^+ = \{i : y_i = 1, A_i = 1\}$ denote their positive-class subsets. We study two fairness objectives, each targeting a different notion of group-level equity.

Demographic Parity (DP) asks whether both groups receive positive predictions at similar rates, regardless of their true outcomes. The DP penalty operationalizes this by penalizing differences in average predicted probability across the two groups:

$$R_{\text{DP}}(\theta) = \left(\frac{1}{|G_0|} \sum_{i \in G_0} \hat{y}_i - \frac{1}{|G_1|} \sum_{i \in G_1} \hat{y}_i \right)^2 \quad (3)$$

Equal Opportunity (EO) asks a narrower question: among individuals who truly belong to the positive class, are both groups detected at similar rates? The EO penalty restricts the same comparison to positive-class samples only, encouraging similar true positive behavior across groups:

$$R_{\text{EO}}(\theta) = \left(\frac{1}{|G_0^+|} \sum_{i \in G_0^+} \hat{y}_i - \frac{1}{|G_1^+|} \sum_{i \in G_1^+} \hat{y}_i \right)^2 \quad (4)$$

These formulations follow Zafar et al. (2017) and Hardt, Price, and Srebro (2016), respectively, building on the fairness regularization paradigm established by Kamishima et al. (2012). In both cases, the squared difference ensures the penalty is non-negative and differentiable everywhere, making it directly compatible with gradient-based optimization rather than requiring a constrained or post-hoc adjustment of the decision rule.

The fairness-aware training objective combines the predictive loss with a single fairness penalty term controlled by a scalar regularization parameter $\lambda \geq 0$:

$$L_{\text{total}}(\theta) = L_{\text{pred}}(\theta) + \lambda \cdot R_{\text{fairness}}(\theta) \quad (5)$$

where R_{fairness} is either R_{DP} or R_{EO} depending on the fairness objective under study. The two objectives are analyzed separately, never jointly, to isolate their individual effects on model behavior and optimization dynamics. When $\lambda = 0$, the objective reduces to standard logistic regression; as λ increases, the optimizer places progressively greater emphasis on the fairness penalty, potentially altering the trade-off between predictive performance and group-level disparity.

To make this trade-off concrete at the gradient level, we expand the gradient of each penalty term with respect to the model parameters. Writing $\hat{y}_i = \sigma(z_i)$ with $z_i = \mathbf{w}^T \mathbf{x}_i + b$,

and using $\sigma'(z_i) = \hat{y}_i(1 - \hat{y}_i)$, the gradient of the DP penalty with respect to $\mathbf{w}$ is

$$\frac{\partial R_{\text{DP}}}{\partial \mathbf{w}} = 2(\bar{y}_{G_0} - \bar{y}_{G_1}) \left(\frac{1}{|G_0|} \sum_{i \in G_0} \hat{y}_i(1 - \hat{y}_i) \mathbf{x}_i - \frac{1}{|G_1|} \sum_{i \in G_1} \hat{y}_i(1 - \hat{y}_i) \mathbf{x}_i \right) \quad (6)$$

where $\bar{y}_{G_0}$ and $\bar{y}_{G_1}$ denote the group-mean predicted probabilities in Equation 3. The gradient of the EO penalty follows the identical structure, restricted to the positive-class subsets G_0^+ and G_1^+ :

$$\frac{\partial R_{\text{EO}}}{\partial \mathbf{w}} = 2(\bar{y}_{G_0^+} - \bar{y}_{G_1^+}) \left(\frac{1}{|G_0^+|} \sum_{i \in G_0^+} \hat{y}_i(1 - \hat{y}_i) \mathbf{x}_i - \frac{1}{|G_1^+|} \sum_{i \in G_1^+} \hat{y}_i(1 - \hat{y}_i) \mathbf{x}_i \right) \quad (7)$$

Two properties of Equations 6 and 7 are worth making explicit, since they motivate the diagnostic framework introduced in the next section. First, each gradient is scaled by the leading factor $(\bar{y}_{G_0} - \bar{y}_{G_1})$ or $(\bar{y}_{G_0^+} - \bar{y}_{G_1^+})$: when the two groups already receive nearly identical predictions, this factor is close to zero and the fairness gradient vanishes regardless of how large λ is. Second, the EO gradient in Equation 7 is computed exclusively from samples in G_0^+ and G_1^+ ; if either subset is small, the gradient is an average over very few terms and is therefore noisier and weaker in magnitude than a gradient averaged over the full dataset. These two observations are precisely what the Base Rate Disparity and Positive Class Sparsity diagnostics introduced below are designed to capture before any training takes place.

The central empirical question motivating this work is whether the effectiveness of fairness-aware optimization depends systematically on structural properties of the dataset itself.

3.2 Dataset Structural Diagnostics

A central observation motivating this work is that the same fairness-aware loss function can produce qualitatively different outcomes across datasets: finding a viable accuracy-fairness operating region on some datasets, failing to reduce disparity on others, and proving unnecessary on datasets where the baseline model is already relatively equitable. We hypothesize that this heterogeneity is not arbitrary, but is influenced by measurable structural properties of the dataset that can be computed prior to model training.

We introduce two such properties as pre-training diagnostic quantities.

Base Rate Disparity (BRD) measures the absolute difference in positive-class rates between the two demographic groups:

$$BRD = \left| \frac{1}{|G_1|} \sum_{i \in G_1} y_i - \frac{1}{|G_0|} \sum_{i \in G_0} y_i \right| \quad (8)$$

BRD captures the extent to which the prediction objective may encode differences in observed outcome distributions between groups. Returning to Equation 6, BRD is closely related to the leading factor $(\bar{y}_{G_0} - \bar{y}_{G_1})$ that scales the DP gradient: a large BRD indicates that the two groups have genuinely different positive-class rates in the training data, which the unconstrained model will tend to reproduce in its predictions. When BRD is large, demographic parity regularization therefore introduces real tension between the predictive objective and the fairness objective: the model is encouraged to preserve predictive accuracy, which is well-served by reproducing the observed group difference, while simultaneously being pushed to reduce that same group-level disparity. Larger BRD values may therefore increase the trade-off between predictive performance and demographic parity objectives, requiring stronger regularization before meaningful fairness improvement can occur. In such settings, increasingly strong regularization may substantially degrade predictive performance before meaningful disparity reduction is achieved.

Positive Class Sparsity (PCS) measures the proportion of minority-group positive-class samples in the full dataset:

$$PCS = \frac{|\{i : y_i = 1, A_i = 0\}|}{N} \quad (9)$$

PCS directly influences the gradient signal available to the EO regularization term. As shown in Equation 7, the EO penalty is computed exclusively over positive-class instances, G_0^+ and G_1^+ , meaning the EO gradient update at each training step is derived only from these samples rather than from the full dataset of size N . When PCS is very small, the minority positive-class group G_0^+ contains few samples, so the average in Equation 7 is taken over a small and disproportionately noisy set, and the resulting fairness gradient is comparatively weak relative to the prediction gradient in Equation 2, which is always averaged over all N samples. Under these conditions, EO regularization struggles to exert meaningful influence during optimization no matter how large λ is set, because the signal it is built on is too sparse to compete.

Worked Example: Heart Disease To make these definitions concrete, we compute BRD and PCS for the UCI Heart Disease dataset before any model is trained. Among male patients, 55.7% exhibit heart disease; among female patients, 26.0% do. Substituting these group-level rates into Equation 8 gives

$$BRD = |0.557 - 0.260| = 0.297 \quad (10)$$

a difference of nearly 30 percentage points between the two groups, the largest observed across the five datasets studied in this work (Table 11). Separately, computing the proportion of female patients with confirmed heart disease relative to the full sample yields

$$PCS = \frac{|\{i : y_i = 1, A_i = 0\}|}{N} = 0.084 \quad (11)$$

placing the EO gradient signal in a moderate range, neither vanishingly small nor especially abundant. As discussed in the Adaptive Fairness Weighting section below, these two values jointly determine how much regularization pressure the adaptive loss function applies to each fairness objective on this dataset, and we return to this same worked example there to show the calculation through to its conclusion.

Together, BRD and PCS form a two-dimensional diagnostic profile for each dataset. Table 11 reports these quantities for all datasets studied in this work alongside the corresponding pre-training diagnostic interpretation.

Three operating regimes emerge from this profile. In the *viable* regime, BRD is moderate to high and PCS is sufficiently large for the fairness gradient to meaningfully influence optimization. In the *ineffective* regime, PCS is extremely low or BRD is large enough that fairness regularization produces only limited improvement relative to its predictive cost. In the *unnecessary* regime, BRD and baseline disparity are already small, indicating that additional regularization may introduce optimization noise without substantial fairness benefit.

These diagnostic interpretations are computed entirely from dataset statistics prior to model training. We do not treat BRD and PCS as universal predictors of fairness outcomes, but rather as empirically motivated indicators intended to characterize important optimization conditions across datasets. In the Experiments and Results section, we evaluate whether these pre-training diagnostics align with the observed experimental behavior across the datasets studied in this work.

3.3 Adaptive Fairness Weighting

We propose an adaptive fairness-aware loss function that incorporates the diagnostic properties directly into the training objective, automatically scaling regularization strength according to dataset conditions. Rather than using a fixed λ , the adaptive objective modulates regularization through dataset-specific weights derived from BRD and PCS:

$$L_{\text{total}}^{DP}(\theta) = L_{\text{pred}}(\theta) + \lambda \cdot w_{BRD} \cdot R_{DP}(\theta) \quad (12)$$

$$L_{\text{total}}^{EO}(\theta) = L_{\text{pred}}(\theta) + \lambda \cdot w_{PCS} \cdot R_{EO}(\theta) \quad (13)$$

where $w_{BRD} \in [0, 1]$ scales the DP penalty according to BRD and $w_{PCS} \in [0, 1]$ scales the EO penalty according to PCS. Under this formulation, λ represents a maximum regularization budget that the practitioner is willing to spend; the weights automatically reduce the *effective* regularization actually applied under dataset conditions associated with weak fairness gradients or limited fairness benefit, without requiring the practitioner to know in advance which regime a given dataset falls into.

Adaptive Weight Design. We adopt piecewise-linear weight formulations, with thresholds empirically derived from the operating regimes identified across the five evaluated datasets:

$$w_{BRD}(BRD) = \begin{cases} 0 & BRD < 0.05 \\ \frac{BRD - 0.05}{0.20} & 0.05 \leq BRD \leq 0.25 \\ 1 & BRD > 0.25 \end{cases} \quad (14)$$

$$w_{PCS}(PCS) = \begin{cases} 0 & PCS < 0.05 \\ \frac{PCS - 0.05}{0.10} & 0.05 \leq PCS \leq 0.15 \\ 1 & PCS > 0.15 \end{cases} \quad (15)$$

The lower thresholds ($BRD = 0.05$, $PCS = 0.05$) correspond to regimes where baseline disparity or gradient signal is too weak to warrant regularization, and the weight resolves to zero, switching the corresponding penalty off entirely. The upper thresholds ($BRD = 0.25$, $PCS = 0.15$) mark regimes where the diagnostic evidence is strong enough that full regularization is warranted, and the weight saturates at one. Between the two thresholds, the weight scales linearly, providing a gradual transition rather than a hard cutoff.

Worked Example: Heart Disease, Continued Returning to the Heart Disease diagnostics computed in Equation 10, we now evaluate both weight functions explicitly. Since $BRD = 0.297$ exceeds the upper threshold of 0.25 in Equation 14,

$$w_{BRD} = 1.0 \quad (16)$$

and the adaptive DP loss in Equation 12 applies the full regularization budget λ , with no attenuation. For PCS, 0.084 falls inside the linear transition region of Equation 15, so

$$w_{PCS} = \frac{0.084 - 0.05}{0.10} = \frac{0.034}{0.10} = 0.342 \quad (17)$$

meaning the adaptive EO loss in Equation 13 applies only 34.2% of the practitioner-specified budget. With the representative budget of $\lambda = 2$ used throughout our experiments, this corresponds to an effective regularization strength of $\lambda \cdot w_{PCS} = 0.684$ for EO, while DP receives the full $\lambda = 2$. This single calculation illustrates the core behavior of the adaptive framework: the two fairness objectives are scaled independently and can diverge substantially in the amount of pressure actually applied, even on the same dataset, because BRD and PCS capture different and largely independent structural properties. We revisit this same numerical result in the Experiments and Results section, where it is compared directly against the outcome of standard fixed- λ regularization on Heart Disease.

Piecewise-linear formulations were chosen over Gaussian or sigmoid alternatives primarily for interpretability. With only five datasets available for threshold calibration, fitting a continuous function with additional shape parameters would risk overfitting the diagnostic framework itself, tuning the weighting curve to the idiosyncrasies of five specific

datasets rather than to a generalizable structural relationship. A piecewise-linear function, by contrast, is fully specified by four numbers, the two thresholds and the (fixed) slope and intercept of the linear segment, and each of those numbers maps directly onto an observed regime boundary in the data. The weighting functions are therefore intentionally simple and interpretable rather than theoretically optimal; we revisit this design choice, and its limitations at the threshold boundaries, in the Discussion and Limitations sections.

When $w_{BRD} = 1$ and $w_{PCS} = 1$, the adaptive objective in Equations 12 and 13 becomes identical to standard fixed- λ regularization, preserving the original fairness formulations of Zafar et al. (2017) and Hardt, Price, and Srebro (2016) exactly. The adaptive framework is therefore best understood as a strict generalization of standard fairness regularization, one that recovers the standard behavior when the diagnostic evidence supports it, and suppresses regularization only when the diagnostic evidence indicates it would be unproductive.

3.4 Experimental Setup

Datasets. We evaluate the proposed framework across five publicly available datasets spanning healthcare and non-healthcare domains. Three healthcare datasets are included as the primary application domain: the *UCI Heart Disease* dataset (303 samples, binary classification of heart disease presence), the *Diabetes 130-US Hospitals* dataset (95,672 samples after preprocessing, 30-day readmission prediction), and the *Kaggle Stroke Prediction* dataset (5,109 samples, binary stroke occurrence prediction). Two non-healthcare benchmark datasets are included to examine whether the observed optimization behaviors generalize across domains: the *UCI Adult Income* dataset (45,222 samples, income classification) and the *UCI German Credit* dataset (1,000 samples, credit risk prediction). In all cases, sex or gender is used as the sensitive attribute, with $A = 0$ and $A = 1$ representing the two demographic groups under consideration.

Preprocessing. Each dataset undergoes standard preprocessing including removal of missing values, one-hot encoding of categorical variables, and `StandardScaler` normalization of continuous features. Several datasets required dataset-specific handling beyond this standard pipeline. The Diabetes 130-US dataset originally expands to 2,303 features once the `diag_1`, `diag_2`, and `diag_3` ICD diagnosis code columns are one-hot encoded; because these high-cardinality columns are also leakage-prone with respect to the readmission outcome, they were dropped, reducing the feature space to 73 columns. The Stroke dataset required median imputation for 201 missing BMI values (3.9% of the sample) and the introduction of an explicit “Unknown” category for 1,544 missing `smoking_status` entries, rather than discarding these records. The German Credit dataset encodes sex within a composite `Attribute9` (personal status) field rather than as a standalone column; codes A92 and A95 were mapped to female and A91, A93, and A94 to male, after which 45 features remained following one-hot encoding of the remaining categorical attributes. All datasets are split into training and test sets using an 80/20 stratified

split. The Adult Income dataset additionally employs a validation set for learning-rate selection due to its substantially larger sample size.

Class Imbalance Handling. Two datasets exhibit substantial class imbalance requiring explicit correction. The Stroke dataset has a 4.9% positive rate and the Diabetes 130-US dataset has an 11.5% positive rate. Class imbalance is a well-documented challenge in clinical prediction tasks requiring explicit correction during training (He and Garcia 2009). For both datasets, we apply inverse-frequency class weighting to the training loss, assigning each sample a weight inversely proportional to its class frequency:

$$w_i = \frac{N}{2 \cdot N_{y_i}} \quad (18)$$

where N is the total number of training samples and N_{y_i} is the number of samples belonging to class y_i . This correction increases the contribution of minority-class samples during optimization, discouraging trivial majority-class prediction behavior. For these datasets, AUC is reported as the primary performance metric rather than accuracy, since accuracy alone can be misleading under severe imbalance. For example, a model predicting only the majority class achieves approximately 88.5% accuracy on the Diabetes dataset without learning any meaningful predictive structure. We return to this distinction in detail in the Results section, where class weighting is shown to introduce its own metric artifacts independent of genuine group disparity.

Model and Optimization. All experiments use logistic regression trained via full-batch gradient descent implemented from scratch, rather than relying on a library implementation, so that the fairness-penalty gradients in Equations 6 and 7 could be computed and verified analytically alongside the standard cross-entropy gradient. The combined parameter update at each training step follows directly from Equation 5:

$$\frac{\partial L_{\text{total}}}{\partial \theta} = \frac{\partial L_{\text{pred}}}{\partial \theta} + \lambda \cdot w \cdot \frac{\partial R_{\text{fairness}}}{\partial \theta} \quad (19)$$

where w is either w_{BRD} or w_{PCS} depending on the fairness objective, and is fixed at $w = 1$ for the standard (non-adaptive) configurations described below. Learning rates and epoch budgets were selected empirically through convergence analysis on the training loss for each dataset, to ensure stable optimization without unnecessary overtraining; the selected values are reported alongside each dataset in Table 11.

Evaluation Protocol. For each dataset, we train three model variants per fairness objective: a baseline model ($\lambda = 0$), a standard fairness-aware model with fixed $\lambda = 2$, and an adaptive fairness-aware model using the same base $\lambda = 2$ scaled by the dataset-specific adaptive weight from Equation 14 or 15. The value $\lambda = 2$ was selected as a representative intermediate regularization regime based on preliminary convergence analysis across datasets, large enough to exert meaningful fairness pressure where the diagnostic evidence supports it, but well short of the over-regularized collapse observed at $\lambda \geq 5$ on several datasets. Each model is evaluated on the held-out test set using four metrics: accuracy,

AUC, Demographic Parity difference, and Equal Opportunity difference. AUC provides a threshold-independent measure of discriminative performance that is particularly appropriate for imbalanced clinical prediction tasks (Fawcett 2006). The DP and EO metrics are computed as absolute group-level disparities using binary predictions at a 0.5 decision threshold. Additionally, we report the full λ -sweep from 0 to 10 for both DP and EO objectives on each dataset, to characterize the observed accuracy–fairness trade-off behavior across regularization strengths rather than at a single operating point. We compare against fixed- λ fairness regularization as the most common deployment baseline; comparison against adaptive multi-objective optimization methods remains future work.

4 Experiments and Results

4.1 Heart Disease — Viable DP Operating Regime

The UCI Heart Disease dataset (Cleveland subset, $n = 297$) serves as the reference case for the viable regularization regime. The diagnostic profile reveals $BRD = 0.297$, male patients exhibit disease prevalence of 55.7% compared to 26.0% among female patients, exactly the worked calculation given in Equation 10, and $PCS = 0.084$, placing the dataset firmly within the predicted viable regime.

The baseline DP-objective model achieves accuracy 0.867 and AUC 0.925, with a baseline DP disparity of 0.278. This disparity reflects the observed disease prevalence gap rather than model instability, a meaningful but structurally tractable fairness scenario consistent with the viable regime prediction. The baseline EO-objective model, trained independently, achieves accuracy 0.878 and AUC 0.927, with a baseline EO disparity of 0.219; the small difference between the two baseline runs reflects ordinary training variation rather than a methodological inconsistency, since each fairness objective is trained from a fresh initialization as described in the Methodology section.

Full Lambda Sweep To characterize the accuracy–fairness trade-off across the full range of regularization strength, rather than at a single operating point, Table 1 reports accuracy, AUC, and disparity for both objectives across $\lambda \in \{0, 0.1, 0.5, 1, 1.5, 1.75, 2, 2.25, 2.5, 3, 5, 10\}$, and Figure 1 plots the same data.

Under DP regularization, disparity does not decline monotonically. It increases at $\lambda = 0.5$ from 0.278 to 0.296, a non-monotonic response consistent with weak fairness pressure perturbing the prediction objective before exerting sufficient corrective influence. Disparity then falls through $\lambda = 1$ and $\lambda = 1.5$, before rising a second time at $\lambda = 1.75$ back to 0.296, and only then settling into a sustained reduction. The clearest disparity reduction in the sweep occurs at $\lambda = 2.25$ – 2.5 , where DP falls to 0.213, a 23% reduction relative to baseline, at an accuracy cost of 5.6 percentage points ($0.867 \rightarrow 0.811$). At the standard comparison point of $\lambda = 2$ used throughout the adaptive evaluation below, DP falls more modestly to 0.269, a 3.3% reduction at an accuracy cost of 3.3 percentage points. We report both checkpoints because they illustrate different things: $\lambda = 2$

λ	Acc	AUC	DP Diff	EO Diff	Notes
<i>DP-objective sweep</i>					
0 (Baseline)	0.867	0.925	0.278	0.219	Baseline
0.1	0.867	0.925	0.278	0.219	No effect
0.5	0.878	0.926	0.296	0.188	Non-monotonic - DP worsens
1	0.867	0.926	0.269	0.188	Recovering
1.5	0.844	0.926	0.259	0.188	Improving
1.75	0.844	0.926	0.296	0.156	Second oscillation
2	0.833	0.926	0.269	0.156	Standard operating point
2.25	0.811	0.926	0.213	0.156	Best DP reduction (23%)
2.5	0.811	0.926	0.213	0.156	Plateau
3	0.822	0.926	0.232	0.125	Slight recovery
5	0.811	0.926	0.232	0.063	Diminishing returns
10	0.711	0.924	0.056	0.000	Over-regularized collapse
<i>EO-objective sweep</i>					
0 (Baseline)	0.878	0.927	0.296	0.188	Baseline
0.1–2.25	0.878	0.927	0.296	0.188	Flat - no EO response
2.5	0.867	0.927	0.269	0.188	First shift (DP side effect)
3	0.867	0.927	0.269	0.188	Plateau continues
5	0.822	0.927	0.250	0.188	Accuracy cost, no EO gain
10	0.833	0.926	0.241	0.063	EO finally responds

Table 1: Full λ -sweep for Heart Disease, DP and EO objectives trained and evaluated independently. EO is unresponsive across more than half the sweep range, only beginning to move at $\lambda \geq 5$.

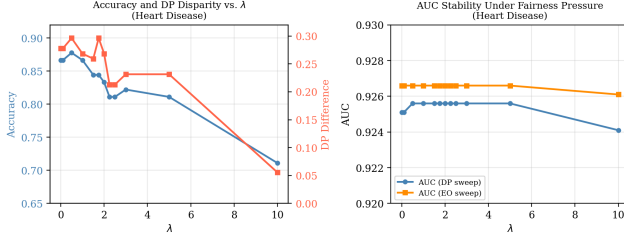


Figure 1: Heart Disease λ -sweep. *Left*: accuracy and DP difference across the DP-objective sweep, showing two non-monotonic oscillations (at $\lambda = 0.5$ and $\lambda = 1.75$) before disparity reduction stabilizes near $\lambda = 2.25$. *Right*: AUC remains essentially flat across both sweeps until $\lambda = 10$, where the DP-objective model shows the first measurable decline.

is the value used consistently across all five datasets for the standard-versus-adaptive comparison, while $\lambda = 2.25$ – 2.5 shows the strongest disparity reduction the sweep actually reaches. AUC remains stable at 0.925–0.926 throughout this entire range, indicating that fairness optimization alters the decision boundary without degrading discriminative capacity; only extreme regularization at $\lambda = 10$ produces a noticeable AUC decline, to 0.924.

EO regularization exhibits a structurally different behavior, visible directly in Table 1: the EO objective is completely flat from $\lambda = 0$ through $\lambda = 2.25$, with accuracy, AUC, DP, and EO all identical across that entire range. Baseline analysis reveals why. $TPR_{Female} = 1.000$ already exceeds $TPR_{Male} = 0.813$ at $\lambda = 0$, meaning the minority subgroup achieves perfect positive-case detection prior to any fairness intervention. Consequently, EO disparity at baseline is driven by lower majority-group TPR rather than

Model	Acc	AUC	DP Diff	EO Diff	Acc Loss
Baseline ($\lambda = 0$)	0.867	0.925	0.278	0.219	+0.000
Standard DP ($\lambda = 2$)	0.833	0.926	0.269	0.156	+0.033
Adaptive DP ($w_{BRD} = 1.0$)	0.833	0.926	0.269	0.156	+0.033
Standard EO ($\lambda = 2$)	0.878	0.927	0.296	0.188	−0.011
Adaptive EO ($w_{PCS} = 0.342$)	0.878	0.927	0.296	0.188	−0.011

Table 2: Heart Disease adaptive comparison. Negative Acc Loss indicates accuracy above the DP-objective baseline, reflecting the small difference between the two independently trained baseline runs discussed above.

minority-group underperformance, and pushing the minority group’s already-saturated TPR higher has no further effect: there is nothing left to correct in that direction. The EO penalty only begins to move the outcome once λ reaches 5 and then 10, where EO disparity finally drops from 0.188 to 0.063, but at a steep accuracy cost and via a different mechanism, namely TPR_{Male} being pulled upward rather than TPR_{Female} being pulled down.

This behavior highlights a limitation of PCS as a standalone EO diagnostic: it predicts gradient signal availability but not the directionality of baseline subgroup TPR geometry, an important direction for future refinement, which we revisit in the Discussion and Limitations sections.

Adaptive Comparison Table 2 reports the standard evaluation protocol described in the Methodology: baseline, standard fixed- $\lambda = 2$ regularization, and adaptive regularization, for both objectives.

Since $BRD = 0.297 > 0.25$, the adaptive weight $w_{BRD} = 1.0$ applies the full DP budget without attenuation, exactly as computed in the worked example in the Methodology section, and produces results identical to the standard formulation: Adaptive DP and Standard DP agree to four decimal places. For EO, $w_{PCS} = 0.342$ reduces the effective regularization budget to $\lambda_{EO} = 0.684$, as derived in Equation 15. Because the EO objective is flat across the entire range from $\lambda = 0$ to $\lambda = 2.25$ on this dataset (Table 1), reducing the effective λ from 2.0 to 0.684 produces no observable change in outcome here: Adaptive EO and Standard EO agree exactly. This is a meaningful result in its own right, though a more modest one than a fixed accuracy-savings figure would suggest: rather than recovering lost accuracy, the adaptive weight correctly recognizes that no amount of pressure within this range would move the EO objective, and avoids spending regularization budget where Table 1 shows none would have been productive regardless. The practical value of the adaptive mechanism on this dataset is therefore one of correct identification rather than measurable accuracy recovery, a distinction that becomes more visible on datasets such as Adult Income and Stroke, discussed next, where standard regularization does measurably waste accuracy that the adaptive formulation recovers.

4.2 Adult Income — Severe Optimization Conflict

Among all datasets examined in this study, Adult Income presents the most structurally adversarial conditions for fairness-aware optimization. With a Base Rate Disparity of

λ	Acc	AUC	DP Diff	EO Diff	Notes
<i>DP-objective sweep</i>					
0 (Baseline)	0.839	0.897	0.176	0.093	Baseline
0.01	0.839	0.898	0.177	0.097	DP worsening begins
0.05	0.838	0.898	0.180	0.091	Monotonic degradation
0.1	0.838	0.898	0.186	0.099	Worsening continues
0.5	0.832	0.900	0.229	0.083	No recovery
1	0.804	0.899	0.294	0.074	Near-collapse
2	0.657	0.895	0.307	0.012	Collapse - 21.6% acc drop
5	0.436	0.882	0.084	0.003	Model broken
10	0.375	0.880	0.057	0.004	Near-random
<i>EO-objective sweep</i>					
0 (Baseline)	0.839	0.897	0.176	0.093	Baseline
0.1	0.838	0.898	0.179	0.093	EO flat, DP creeping
0.5	0.839	0.899	0.196	0.096	EO unresponsive
1	0.833	0.899	0.216	0.102	DP worsening; EO static
2	0.829	0.900	0.237	0.075	EO slight gain; DP worse
10	0.677	0.895	0.325	0.029	Objective conflict dominates

Table 3: Full λ -sweep for Adult Income, DP and EO objectives trained independently. DP worsens monotonically from $\lambda = 0.005$ onward before catastrophic collapse at $\lambda = 2$. EO regularization causes DP to worsen across the entire sweep, confirming structural metric conflict.

$BRD = 0.199$, the dataset exhibits a pronounced income gap between demographic groups: male subjects earn above \$50K at a rate of 31.3% compared to 11.4% among female subjects. Positive Class Sparsity is $PCS = 0.037$, well below the 0.05 threshold at which EO gradient signal becomes effective. The diagnostic framework places this dataset firmly in the predicted ineffective regime: BRD is large enough that the prediction gradient strongly encodes observed disparities present in the training distribution, and PCS is insufficient for EO regularization to exert meaningful influence. Crucially, the sensitive attribute was excluded from the feature matrix prior to training, ensuring that the observed disparities reflect patterns the model learned from socioeconomic features rather than direct demographic conditioning.

The baseline model achieves accuracy 0.839 and AUC 0.897, with a baseline DP difference of 0.176 and EO difference of 0.093. Unlike Heart Disease, where the baseline disparity was large but structurally tractable, Adult Income presents a baseline where the disparity is deeply embedded in the feature correlations learned during training: male and female subjects differ substantially in occupation, education level, hours worked, and capital gains, all of which the model has accurately learned. The conditions for optimization conflict are therefore fully established before regularization even begins.

Full Lambda Sweep Table 3 reports the full λ sweep for both objectives, and Figure 2 visualizes the DP trade-off frontier and AUC stability.

What follows under DP regularization is not a gradual trade-off but a rapid and sustained intensification of optimization conflict. The fine-grained search at $\lambda \in \{0.001, 0.005, 0.01, 0.02, 0.05\}$ confirms that DP disparity begins worsening at $\lambda = 0.005$ and increases monotonically:

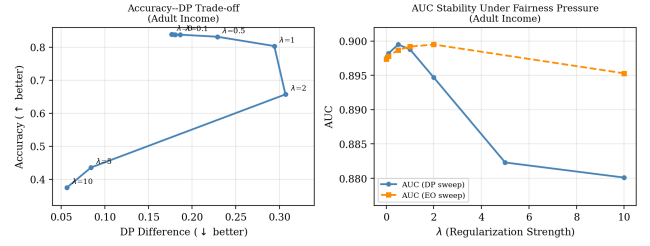


Figure 2: Adult Income regularization behavior. *Left*: Accuracy–DP trade-off frontier across the full λ sweep. The curve traces rightward (disparity worsening) before dropping sharply at $\lambda = 2$, with no intermediate point offering meaningful fairness improvement at acceptable accuracy cost, no viable operating region exists. *Right*: AUC stability under both DP and EO sweeps. AUC remains above 0.880 even at $\lambda = 10$ where accuracy collapses to 0.375, demonstrating that discriminative capacity is preserved even as the decision boundary deteriorates.

cally: no evaluated λ produces meaningful DP improvement without considerable predictive degradation. At $\lambda = 0.1$, DP disparity has already increased from 0.176 to 0.186. At $\lambda = 0.5$, it reaches 0.229. At $\lambda = 1$, the model approaches near-collapse with accuracy at 0.804 and DP at 0.294. By $\lambda = 2$, accuracy has dropped 21.6% to 0.657 while DP has actually worsened to 0.307, higher than the baseline it was supposed to reduce.

This monotonic worsening distinguishes Adult Income sharply from Heart Disease. Whereas Heart Disease exhibited a single transient increase in disparity at $\lambda = 0.5$ before recovering, Adult Income shows no recovery at any evaluated λ value through $\lambda = 2$. The trade-off frontier in Figure 2 makes this visible: rather than tracing a southwest arc toward low-disparity, high-accuracy outcomes, it moves east (worsening disparity) before collapsing south (accuracy loss), with no viable operating point along the path.

The mechanism is precisely what the BRD diagnostic predicts. With $BRD = 0.199$, the prediction gradient encodes a real and pervasive structural difference in income rates between male and female subjects. When DP regularization asks the optimizer to equalize predicted positive rates, it is asking the model to override a signal embedded across occupation, education, capital gains, and hours-worked features simultaneously. The resulting gradient conflict is not a numerical instability but a fundamental tension between two legitimate but structurally incompatible objectives on this data, consistent with theoretical characterizations of empirical risk minimization under fairness constraints (Donini et al. 2018). This is the quantitative expression of what Berk et al. (2017) described qualitatively as domain-dependent trade-offs lacking prescriptive universals; the BRD diagnostic provides the structural explanation they could not.

EO regularization reveals a different but equally fundamental problem: metric conflict. As shown in Table 3, optimizing the EO objective causes DP disparity to worsen monotonically, from 0.176 at baseline to 0.325 at $\lambda = 10$. Conversely, optimizing the DP objective reduces EO difference from 0.093 to 0.012 at $\lambda = 2$, as the collapsing model

Model	Acc	AUC	DP Diff	EO Diff	Acc Loss
Baseline ($\lambda = 0$)	0.839	0.897	0.176	0.093	+0.000
Standard DP ($\lambda = 2$)	0.657	0.895	0.307	0.012	+0.182
Adaptive DP ($w_{BRD} = 0.745$)	0.733	0.897	0.330	0.052	+0.106
Standard EO ($\lambda = 2$)	0.829	0.900	0.237	0.075	+0.009
Adaptive EO ($w_{PCS} = 0.000$)	0.839	0.897	0.176	0.093	+0.000

Table 4: Adult Income adaptive comparison. Adaptive DP recovers 41.6% of the accuracy loss caused by standard regularization. Adaptive EO switches off entirely ($w_{PCS} = 0$), avoiding 100% of the EO accuracy cost and preventing the DP worsening that standard EO introduces.

begins predicting uniformly. The two metrics are actively competing for the same gradient budget on this dataset, a structural property determined by the relationship between group sizes, base rates, and positive class distributions.

A notable cross-cutting observation visible in Figure 2 is AUC resilience. AUC remains above 0.880 across the entire DP sweep, including at $\lambda = 10$ where accuracy has fallen to 0.375, near-random prediction. This confirms that the model retains its underlying discriminative capacity even as the decision boundary deteriorates under extreme fairness pressure, a clinically relevant property discussed further in the Discussion section.

Adaptive Comparison Table 4 reports the standard comparison protocol.

Since $BRD = 0.199$ falls in the linear transition region of Equation 14, the adaptive weight is

$$w_{BRD} = \frac{0.199 - 0.05}{0.20} = \frac{0.149}{0.20} = 0.745 \quad (20)$$

reducing the effective DP regularization to $\lambda_{DP} = 2.0 \times 0.745 = 1.489$, below the collapse threshold observed at $\lambda = 2$. The result is a recovery of 41.6% of the accuracy loss: accuracy improves from 0.657 under standard $\lambda = 2$ to 0.733 under adaptive $\lambda = 1.489$. Note that adaptive DP does not resolve the underlying structural conflict: DP disparity still worsens (from 0.176 to 0.330), and no viable fairness operating point is recovered. The adaptive framework is not designed to fix a structurally unfavorable dataset; it is designed to moderate the collateral damage of applying pressure where none can be productively absorbed.

For EO, $PCS = 0.037 < 0.05$ places the adaptive weight exactly at zero: $w_{PCS} = 0.000$, and Adaptive EO is identical to the baseline. This prevents the 0.93% accuracy degradation that standard EO at $\lambda = 2$ introduces, and more importantly, it prevents the DP worsening from 0.176 to 0.237 that standard EO causes as a side effect of metric conflict. The adaptive framework correctly identifies that on a dataset where optimizing EO structurally worsens DP, applying zero EO pressure is preferable to applying any positive pressure.

The result here is therefore not a success story for fairness improvement; it is a demonstration that a principled diagnostic framework can substantially reduce the collateral damage of misapplied regularization, even on datasets where no viable fairness operating point exists.

4.3 Diabetes 130-US — Unnecessary Regularization and Class Imbalance Artifacts

The Diabetes 130-US dataset demonstrates two related but distinct phenomena that arise when fairness regularization is applied under severe class imbalance: the diagnostic framework correctly identifies DP regularization as structurally unnecessary, and the dataset exposes a practically important interaction between class imbalance correction and fairness metric interpretation that extends beyond the regularization setting studied here.

With $BRD = 0.0014$, male and female patients have near-identical 30-day readmission rates of 11.45% and 11.59% respectively, the smallest base rate gap in this study by a wide margin. $PCS = 0.063$ places the EO gradient signal in the moderate range. The adaptive framework resolves $w_{BRD} = 0.000$ and $w_{PCS} = 0.125$, switching DP regularization off entirely and applying only 12.5% of the EO budget.

Class Imbalance Artifact The dataset has an 11.5% positive readmission rate, requiring inverse-frequency class weighting to prevent the model from trivially predicting the majority class. Without class weighting, the unweighted baseline achieves 0.885 accuracy by predicting the majority class for nearly all instances, yielding near-zero DP and EO differences: apparent fairness achieved through prediction collapse rather than genuine equitable behavior. Applying class weighting restores meaningful discriminative learning, but simultaneously introduces apparent fairness disparities of $DP = 0.032$ and $EO = 0.025$ despite near-identical male and female readmission rates. These disparities are not genuine group inequities; they are artifacts of how the weighting correction interacts with the prediction distribution across groups. A practitioner who observes these values without understanding their origin and applies fairness regularization in response is optimizing against a measurement artifact.

Full Lambda Sweep Table 5 reports the full λ sweep and Figure 3 visualizes the central finding: accuracy and AUC diverge monotonically under increasing regularization pressure.

As Figure 3 shows, accuracy climbs monotonically from 0.669 at $\lambda = 0$ to 0.813 at $\lambda = 10$ while AUC falls from 0.651 to 0.649. This divergence is the central empirical observation for this dataset and the reason AUC must be treated as the primary evaluation metric under class imbalance. What is happening mechanically is straightforward: increasing λ pushes the model to equalize predicted positive rates across groups, but on a dataset where nearly 89% of samples are negative, the most effective way to satisfy a low-prediction-rate constraint is to predict fewer positives overall, which means predicting the majority class more frequently. Accuracy rises because the majority-class prediction happens to be correct more often on an imbalanced test set. AUC falls because the model’s ability to rank patients by true risk is being systematically degraded. A deployment decision based on the accuracy column alone would conclude that $\lambda = 10$ produces the best model; the AUC column reveals the opposite.

λ	Acc	AUC	DP Diff	EO Diff	Notes
<i>DP-objective sweep</i>					
0 (Baseline)	0.669	0.651	0.032	0.025	Weighted baseline
0.1	0.671	0.651	0.033	0.027	Acc rising; AUC flat
0.5	0.679	0.651	0.033	0.026	Trend established
1	0.689	0.651	0.033	0.029	Model reverting toward majority
2	0.709	0.650	0.028	0.019	Standard operating point
3	0.727	0.650	0.030	0.024	AUC degrading
5	0.758	0.650	0.021	0.018	Apparent accuracy gain
10	0.813	0.649	0.018	0.029	Near-majority prediction
<i>EO-objective sweep</i>					
0 (Baseline)	0.669	0.651	0.032	0.025	Weighted baseline
0.5	0.679	0.651	0.033	0.027	Minimal response
1	0.689	0.651	0.033	0.028	Flat
2	0.709	0.650	0.028	0.021	Standard operating point
5	0.757	0.650	0.021	0.015	Same pattern as DP
10	0.814	0.649	0.019	0.029	Collapse mirrors DP

Table 5: Full λ -sweep for Diabetes 130-US, DP and EO objectives trained independently using class-weighted loss. Accuracy rises monotonically as λ increases while AUC falls, the opposite of what genuine fairness improvement looks like. Both metrics must be reported together on imbalanced datasets.

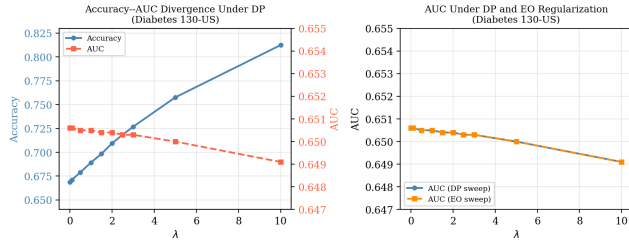


Figure 3: Diabetes 130-US regularization behavior. *Left*: Accuracy and AUC under increasing DP regularization. Accuracy rises from 0.669 at $\lambda = 0$ to 0.813 at $\lambda = 10$ while AUC falls from 0.651 to 0.649, a divergence that reveals the model is reverting toward majority-class prediction rather than learning a better decision boundary. *Right*: AUC under both DP and EO sweeps, showing near-identical behavior consistent with both objectives pushing the model in the same direction on this dataset.

The EO sweep (Table 5, Figure 3 right) mirrors the DP pattern almost exactly, confirming that both objectives are activating the same underlying mechanism on this dataset: reverting toward majority-class prediction under fairness pressure, rather than genuinely redistributing predictive accuracy across groups. The near-identical AUC trajectories under both objectives suggest that the class weighting artifact dominates the optimization dynamics regardless of which fairness criterion is applied.

Adaptive Comparison Table 6 reports the standard three-model comparison.

Since $BRD = 0.0014 < 0.05$, the adaptive weight resolves to $w_{BRD} = 0.000$ and Adaptive DP is identical to the baseline: 0.669 accuracy and 0.651 AUC, preserving the baseline AUC entirely. Standard DP at $\lambda = 2$ produces an apparent accuracy gain of 4.05 percentage points (0.669 $\rightarrow$ 0.709), but this conceals an AUC decline from

Model	Acc	AUC	DP Diff	EO Diff	Acc Loss
Baseline ($\lambda = 0$)	0.669	0.651	0.032	0.025	+0.000
Standard DP ($\lambda = 2$)	0.709	0.650	0.028	0.019	-0.041
Adaptive DP ($w_{BRD} = 0.000$)	0.669	0.651	0.032	0.025	+0.000
Standard EO ($\lambda = 2$)	0.709	0.650	0.028	0.021	-0.040
Adaptive EO ($w_{PCS} = 0.125$)	0.673	0.651	0.032	0.027	-0.005

Table 6: Diabetes 130-US adaptive comparison. Negative Acc Loss values are class imbalance artifacts: accuracy *appears* to improve under regularization but AUC simultaneously declines, revealing model degradation. Adaptive DP preserves baseline AUC entirely by applying zero pressure. Adaptive EO applies a scaled penalty ($\lambda_{EO} = 0.250$) with minimal impact.

0.651 to 0.650. The adaptive framework correctly avoids this outcome by recognizing that $BRD = 0.001$ reflects no genuine group disparity warranting correction.

For EO, $PCS = 0.063$ falls in the linear transition region of Equation 15, giving $w_{PCS} = (0.063 - 0.05)/0.10 = 0.125$ and an effective budget of $\lambda_{EO} = 0.250$. Adaptive EO produces a small accuracy change of -0.005 with negligible EO improvement (0.025 $\rightarrow$ 0.027, a slight worsening consistent with the near-zero underlying disparity), while preserving AUC at 0.651. The scaled penalty at $\lambda = 0.250$ is insufficient to trigger the majority-class reversion visible at higher λ values, keeping the model in the region where accuracy and AUC remain consistent.

The central practical implication of this dataset is direct: on imbalanced clinical prediction tasks, accuracy is not a reliable primary metric for evaluating either predictive performance or the effect of fairness interventions. AUC should be reported alongside, or in preference to, accuracy in such settings. This applies broadly beyond the fairness regularization context studied here; it holds for any intervention that affects the model’s prediction threshold or positive class rate, including class weighting itself.

4.4 Stroke — Clean Unnecessary Regime Validation

The Kaggle Stroke Prediction dataset serves as the cleanest validation case in this study: both adaptive weights resolve exactly to zero, both standard regularization methods cause unnecessary predictive damage, and the adaptive framework prevents all of it. No dataset in the study more directly demonstrates the framework’s core function.

With $BRD = 0.004$, male and female patients exhibit stroke rates of 5.11% and 4.71% respectively, a difference of less than half a percentage point. $PCS = 0.028$, corresponding to only 113 female stroke cases in the full training set, a count too small for the EO gradient to compete meaningfully with the prediction gradient averaged over all N training samples. Both adaptive weights resolve to exactly zero: $w_{BRD} = 0.000$ and $w_{PCS} = 0.000$.

The baseline achieves accuracy 0.730 and AUC 0.839, with a baseline DP difference of 0.023 and EO difference of 0.066. The EO difference at baseline warrants a note: $TPR_{Female} = 0.828$ already exceeds $TPR_{Male} = 0.762$, meaning the minority subgroup achieves higher positive-case detection than the majority group prior to any fairness

λ	Acc	AUC	DP Diff	EO Diff	Notes
<i>DP-objective sweep</i>					
0 (Baseline)	0.730	0.839	0.023	0.066	Baseline
0.1	0.728	0.839	0.020	0.066	Minor degradation
0.5	0.726	0.839	0.017	0.066	Gradual accuracy loss
1	0.722	0.839	0.019	0.018	Oscillating DP
2	0.708	0.840	0.028	0.030	Standard point - 2.15% loss
3	0.698	0.840	0.018	0.030	Continuing loss
5	0.670	0.840	0.031	0.030	No improvement
10	0.580	0.839	0.034	0.030	Severe degradation
<i>EO-objective sweep</i>					
0 (Baseline)	0.730	0.839	0.023	0.066	Baseline
0.1	0.726	0.839	0.017	0.066	EO flat
0.5	0.706	0.841	0.027	0.030	First large accuracy drop
1	0.678	0.840	0.024	0.030	Accelerating loss
2	0.625	0.840	0.039	0.030	10.47% accuracy loss
3	0.572	0.839	0.037	0.030	EO stuck at 0.030
5	0.497	0.837	0.017	0.077	Model degrading
10	0.246	0.835	0.019	0.000	Gaming - TPR=1.0 both groups

Table 7: Full λ -sweep for Stroke, DP and EO objectives trained with class weighting. EO causes far greater accuracy damage than DP at every comparable λ value, while neither objective produces meaningful fairness improvement. At $\lambda = 10$, EO satisfies the metric only through degenerate prediction where both groups achieve $TPR = 1.000$ by predicting everyone as positive.

intervention. The EO disparity is therefore driven by lower majority-group TPR rather than minority-group underperformance, a reversed disparity structure that PCS does not capture and that standard EO regularization is not designed to address in this direction.

Full Lambda Sweep Table 7 reports the full λ sweep and Figure 4 shows the accuracy collapse and AUC stability.

Under DP regularization, accuracy degrades gradually from 0.730 at baseline to 0.580 at $\lambda = 10$, a 15-point drop, while DP disparity oscillates without meaningful improvement and never reaches a value substantially below the baseline of 0.023. The fairness penalty in this setting is extremely weak — the near-absent group disparity signal means the DP gradient is negligible relative to the prediction gradient, so even moderate regularization pressure diverts optimization effort without exerting coherent corrective influence. The result is pure accuracy cost with zero fairness benefit.

EO regularization produces a sharper collapse. Accuracy falls from 0.730 at baseline to 0.625 at $\lambda = 2$ (a 10.47% loss), 0.497 at $\lambda = 5$, and 0.246 at $\lambda = 10$. This 48-point drop is among the steepest in the study. EO difference from $\lambda = 0.5$ through $\lambda = 5$ is locked at approximately 0.030 regardless of how high λ is set, indicating that the EO gradient is too weak to move the metric despite causing severe collateral damage to accuracy. At $\lambda = 10$, EO finally reaches 0.000, but exclusively through degenerate behavior: both groups achieve $TPR = 1.000$ because the model predicts everyone as having a stroke, satisfying the equalization constraint mathematically without achieving any meaningful fairness objective. This is optimization gaming rather than genuine equity.

A notable observation from the AUC panel in Figure 4:

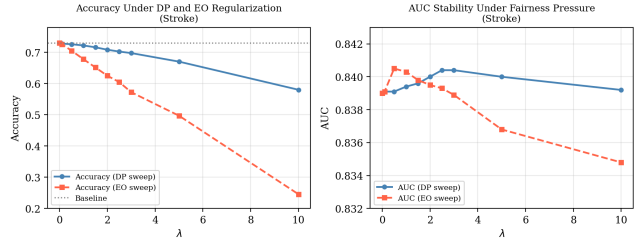


Figure 4: Stroke regularization behavior. *Left*: Accuracy under DP and EO sweeps relative to the baseline (dotted line). EO accuracy collapses from 0.730 at baseline to 0.246 at $\lambda = 10$, a 48-point drop with no genuine fairness improvement. DP degrades more gradually but also produces no operating point with meaningfully reduced disparity. *Right*: AUC remains between 0.835 and 0.841 across both sweeps until extreme regularization, while the DP sweep shows AUC broadly stable. The DP sweep’s AUC mildly improves at moderate λ values as the model explores the decision boundary, even as accuracy falls.

Model	Acc	AUC	DP Diff	EO Diff	Acc Loss
Baseline ($\lambda = 0$)	0.730	0.839	0.023	0.066	+0.000
Standard DP ($\lambda = 2$)	0.708	0.840	0.028	0.030	+0.022
Adaptive DP ($w_{BRD} = 0.000$)	0.730	0.839	0.023	0.066	+0.000
Standard EO ($\lambda = 2$)	0.625	0.840	0.039	0.030	+0.105
Adaptive EO ($w_{PCS} = 0.000$)	0.730	0.839	0.023	0.066	+0.000

Table 8: Stroke adaptive comparison. Both adaptive weights resolve to zero, making Adaptive DP and Adaptive EO identical to the baseline. Standard DP inflicted a 2.15% unnecessary accuracy loss; Standard EO inflicted a 10.47% unnecessary accuracy loss. The adaptive framework prevents 100% of this damage on both objectives.

even as EO accuracy collapses to 0.246 at $\lambda = 10$, AUC remains at 0.835, demonstrating that the model retains discriminative capacity even under extreme and destructive regularization pressure. The DP AUC is essentially flat across the sweep at 0.839–0.840, consistent with the AUC resilience finding observed across all five datasets.

Adaptive Comparison Table 8 reports the standard three-model comparison.

Since both $BRD = 0.004 < 0.05$ and $PCS = 0.028 < 0.05$, both weight functions from Equations 14 and 15 resolve to zero, and both adaptive models are identical to the baseline: 0.730 accuracy, 0.839 AUC, 0.023 DP, 0.066 EO. Standard DP at $\lambda = 2$ inflicted a 2.15 percentage point accuracy loss with no fairness benefit; the adaptive framework saves 100% of that loss. Standard EO at $\lambda = 2$ inflicted a 10.47 percentage point loss, again with no meaningful fairness improvement; the adaptive framework saves 100% of that loss too.

Stroke is the cleanest validation case in this study precisely because neither objective provides any fairness benefit at any λ value while both cause accuracy damage, and both adaptive weights correctly resolve to zero without any dataset-specific tuning. The pre-training diagnostics, computed from BRD and PCS alone before any model training, correctly identified that no regularization was war-

ranted, and the adaptive framework enforced that judgment automatically.

4.5 German Credit — Boundary Conditions and Independent Diagnostic Dimensions

German Credit is the most structurally nuanced dataset in this study. It presents a low-disparity regime where DP regularization is near-unnecessary but EO regularization finds genuine headroom, demonstrating that BRD and PCS characterize independent dimensions of dataset structure that can point in opposite directions simultaneously.

The diagnostic profile reflects this divergence. Base Rate Disparity is $BRD = 0.075$, placing the dataset near the lower boundary of the framework’s sensitivity range. Male subjects hold good credit at a rate of 72.3% compared to 64.8% among female subjects, a gap that is present but modest relative to the other datasets studied here. The sensitive attribute also requires careful extraction: sex is encoded within a composite personal-status attribute (`Attribute9`), with code A92 representing female subjects and A91, A93, and A94 representing male subjects. The fairness analysis is therefore scoped specifically to these subgroup definitions, with the A95 code absent from this particular dataset.

$PCS = 0.201$, the highest value in this study, indicating substantial positive-class minority representation. These two properties diverge in their diagnostic implications: BRD suggests comparatively limited DP necessity, while PCS indicates that EO gradient signal is sufficient for meaningful optimization influence. The adaptive framework reflects this distinction directly, resolving $w_{BRD} = 0.124$ and $w_{PCS} = 1.000$ — a partial DP weight and a full EO weight on the same dataset.

The baseline achieves accuracy 0.710 and AUC 0.752, with Demographic Parity difference of 0.010 and Equal Opportunity difference of 0.030, among the smallest baseline disparities in the study.

Full Lambda Sweep Table 9 and Figure 5 show the full sweep, which makes the divergence between DP and EO behavior visually immediate.

Under DP regularization, disparity oscillates non-directionally across the full sweep with no coherent improvement trajectory. The maximum fairness penalty observed during DP training is 0.00121, approximately 45 times weaker than Heart Disease, reflecting the near-absent group disparity signal at $BRD = 0.075$. At $\lambda = 10$, DP worsens sharply to 0.055 as aggressive regularization perturbs an already equitable decision boundary, confirming that the baseline model is already near-optimal on this metric and that regularization adds noise rather than signal.

EO regularization behaves structurally differently. EO difference is completely flat from $\lambda = 0$ through $\lambda = 1$, then drops from 0.030 to 0.005 at $\lambda = 2$, an 83% reduction at only 0.7% accuracy cost. This improvement is driven by TPR_{Female} increasing from 0.800 to 0.825 while TPR_{Male} remains stable at 0.830, a genuine equalization rather than a collapse artifact. The EO operating point holds through $\lambda = 2.5$ and $\lambda = 3$ before rebounding at higher λ

λ	Acc	AUC	DP Diff	EO Diff	Notes
<i>DP-objective sweep</i>					
0 (Baseline)	0.710	0.752	0.010	0.030	Near-zero DP baseline
0.1	0.710	0.751	0.010	0.030	No effect
0.5	0.705	0.751	0.007	0.005	Noise
1	0.710	0.752	0.014	0.015	Oscillating
2	0.715	0.751	0.021	0.025	DP worsened
2.5	0.725	0.751	0.012	0.010	Non-directional
5	0.720	0.750	0.010	0.035	No coherent trend
10	0.705	0.752	0.055	0.030	DP sharply worsened
<i>EO-objective sweep</i>					
0 (Baseline)	0.710	0.752	0.010	0.030	Baseline
0.1–1	0.710	0.752	0.010	0.030	EO completely flat
1.5	0.700	0.751	0.024	0.030	Pre-threshold noise
2	0.705	0.751	0.007	0.005	EO drops 83%
2.5–3	0.705	0.751	0.007	0.005	Stable plateau
5	0.710	0.752	0.014	0.015	Slight rebound
10	0.715	0.751	0.021	0.025	EO creeping back

Table 9: Full λ -sweep for German Credit. DP regularization produces no coherent improvement trajectory and worsens to 0.055 at $\lambda = 10$. EO regularization is flat from $\lambda = 0$ through $\lambda = 1$, then drops 83% at $\lambda = 2$ and holds, before rebounding at higher λ values.

Model	Acc	AUC	DP Diff	EO Diff	Acc Loss
Baseline ($\lambda = 0$)	0.710	0.752	0.010	0.030	+0.000
Standard DP ($\lambda = 2$)	0.715	0.751	0.021	0.025	−0.005
Adaptive DP ($w_{BRD} = 0.124$)	0.700	0.752	0.024	0.030	+0.010
Standard EO ($\lambda = 2$)	0.705	0.751	0.007	0.005	+0.005
Adaptive EO ($w_{PCS} = 1.000$)	0.705	0.751	0.007	0.005	+0.005

Table 10: German Credit adaptive comparison. Standard DP shows a negative Acc Loss (apparent improvement) reflecting small-sample noise rather than genuine benefit; DP disparity simultaneously worsened. Adaptive EO is identical to Standard EO since $w_{PCS} = 1.000$, correctly applying the full budget. Adaptive DP introduces a small accuracy cost at the boundary edge case where $BRD = 0.075$ is close to the 0.05 threshold.

values, indicating that the EO improvement is real but fragile beyond its optimal range. $PCS = 0.201$ provides the gradient signal for this improvement: with substantial minority positive-class representation, the EO penalty has a large enough sample pool in G_0^+ to exert coherent gradient pressure — unlike on Adult Income or Stroke where PCS was too sparse.

Adaptive Comparison Table 10 reports the standard three-model comparison.

Since $BRD = 0.075$ falls in the linear transition region of Equation 14, the adaptive DP weight is

$$w_{BRD} = \frac{0.075 - 0.05}{0.20} = \frac{0.025}{0.20} = 0.124 \quad (21)$$

giving an effective budget of $\lambda_{DP} = 0.248$. Adaptive DP produces an accuracy cost of +1.0% (0.710 $\rightarrow$ 0.700) with no meaningful DP improvement, confirming the expected behavior for a dataset near the lower BRD threshold. This is also the clearest illustration of a framework limitation: $BRD = 0.075$ is only 0.025 above the lower threshold,

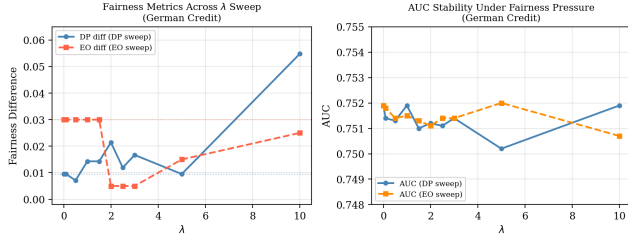


Figure 5: German Credit λ -sweep. *Left*: DP difference (solid blue) oscillates non-directionally around its near-zero baseline, with dotted reference lines marking the starting values. EO difference (dashed red) is flat through $\lambda = 1$ then drops sharply at $\lambda = 2$ before partially rebounding at high λ . The two metrics respond completely differently to the same regularization procedure on the same dataset, directly reflecting the divergence between $BRD = 0.075$ and $PCS = 0.201$. *Right*: AUC remains tightly bounded between 0.750 and 0.752 across both sweeps, consistent with the cross-dataset AUC resilience finding.

meaning the weight function assigns a non-trivial penalty of 12.4% to a dataset that is effectively already fair. German Credit suggests the lower BRD threshold may benefit from upward refinement in future work, a point acknowledged explicitly in the Limitations section.

For EO, $PCS = 0.201 > 0.15$ saturates the weight function at $w_{PCS} = 1.000$, and Adaptive EO is identical to Standard EO at $\lambda = 2$: accuracy 0.705, AUC 0.751, EO 0.005. The adaptive framework correctly applies the full EO budget since PCS provides sufficient gradient signal, and the result is the genuine 83% EO reduction confirmed in the sweep table above.

Standard DP at $\lambda = 2$ shows a negative Acc Loss of -0.005 , an apparent accuracy improvement that merits a brief note. With only 1,000 samples and a near-zero baseline DP disparity, the decision boundary is close to optimal from the start, and small perturbations from regularization can happen to improve boundary placement on the test split rather than degrading it. This is a small-sample noise artifact rather than genuine benefit: the DP column reveals the opposite direction, with disparity worsening from 0.010 to 0.021. Any interpretation of Standard DP as beneficial on German Credit should be anchored in the fairness metrics, not the accuracy column alone.

German Credit isolates a property no previous dataset demonstrated independently: BRD and PCS characterize genuinely independent structural dimensions. Low BRD limits DP utility while high PCS simultaneously enables meaningful EO optimization. The two diagnostics can diverge, and the adaptive framework handles this correctly by scaling each penalty weight from its own structural property rather than applying a single scalar to both objectives.

5 Discussion

5.1 Cross-Dataset Structural Patterns

Taken together, the five datasets reveal that fairness-aware optimization is not a property of the method but of the data. The same gradient-based regularization procedure, applied

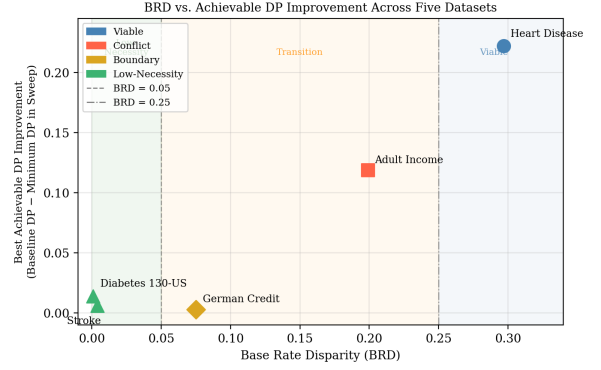


Figure 6: BRD vs. best achievable DP improvement (baseline DP minus minimum DP across the full λ -sweep) across all five datasets. The monotonic relationship between BRD and achievable improvement is visible across the full range, with the two diagnostic threshold lines at $BRD = 0.05$ and $BRD = 0.25$ clearly separating the three operating regimes. Heart Disease (Viable) achieves the highest DP improvement. Adult Income (Conflict) achieves meaningful reduction only at the cost of catastrophic accuracy degradation. German Credit, Stroke, and Diabetes 130-US cluster near zero, consistent with the Low-Necessity and Boundary regime predictions.

with the same hyperparameter, produces qualitatively distinct outcomes depending on the structural conditions of the dataset it encounters. Prior work observed similar heterogeneity across clinical settings (Pföhl, Foryciarz, and Shah 2021) and regression domains (Berk et al. 2017) without identifying a structural explanation. The patterns observed here suggest that BRD and PCS characterize the underlying conditions responsible for this heterogeneity.

Base Rate Disparity emerges as the primary determinant of DP regularization behavior. When BRD is high, the prediction objective reflects substantial group-level disparities, creating a manageable gradient conflict with a viable operating region. At moderate BRD, disparity becomes deeply embedded in learned feature correlations without yielding a favorable optimization geometry, producing sustained conflict. At low BRD, baseline disparities are already small, the prediction gradient encodes minimal group-level signal, and DP regularization introduces noise rather than improvement. This pattern appears consistently across datasets and is visible directly in Figure 6, which plots BRD against the best DP improvement achievable across the full λ -sweep for each dataset.

Positive Class Sparsity operates independently of BRD, determining whether the EO gradient can exert meaningful influence. Below approximately $PCS = 0.05$, the minority positive-class group contributes a near-zero gradient signal and EO regularization produces no coherent improvement regardless of λ . As PCS increases, EO responsiveness grows proportionally, as demonstrated by German Credit achieving an 83% EO reduction at $PCS = 0.201$ while Adult Income and Stroke produce no coherent EO improvement at $PCS = 0.037$ and $PCS = 0.028$ respectively.

Three operating regimes emerge. The *viable* regime is

characterized by BRD sufficient to indicate real group disparity and PCS sufficient to support gradient-based correction. The *conflict* regime is characterized by BRD large enough to embed disparity deeply in learned feature correlations, making correction structurally difficult. The *low-necessity* regime is characterized by BRD too small to warrant DP intervention, though EO may remain responsive where PCS is adequate. German Credit occupies a boundary position, exhibiting low DP necessity alongside meaningful EO responsiveness and demonstrating that BRD and PCS characterize genuinely independent optimization dimensions rather than two facets of the same underlying quantity.

Table 11 summarizes the diagnostic properties, predicted regimes, observed outcomes, and adaptive behavior across all five datasets.

5.2 Adaptive Regularization as Optimization Governance

The conventional framing of adaptive fairness methods positions adaptivity as a mechanism for achieving better fairness outcomes. The results of this study suggest a more precise characterization: adaptive regularization is a mechanism for preventing misaligned fairness pressure, not for maximizing fairness improvement.

This distinction matters. On Heart Disease, where DP regularization is genuinely warranted, the adaptive framework applies the full penalty ($w_{BRD} = 1.0$) and produces results identical to standard regularization; it does not improve on what standard regularization already achieves. On Adult Income, where DP regularization finds no viable operating point, the adaptive framework reduces the effective λ from 2.0 to 1.489, recovering 41.6% of the unnecessary accuracy loss caused by standard regularization, but it does not fix the underlying optimization conflict, and DP disparity remains elevated.

On Stroke, where both BRD and PCS fall below their respective thresholds, the adaptive framework switches off entirely, preventing the 10.47% accuracy loss that standard EO regularization inflicts with zero fairness benefit (Table 8). On Diabetes 130-US, where BRD is near-zero but PCS is moderate, Adaptive DP switches off while Adaptive EO applies a scaled penalty; in both cases the adaptive framework avoids the spurious accuracy gains under regularization that AUC correctly identifies as model degradation, as shown in Figure 3. In no case does the adaptive framework produce a fairness improvement that standard regularization at the same base λ would not also produce if the dataset structure supported it.

The contribution of the adaptive framework is therefore not that it achieves better fairness. It is that it prevents the practitioner from inadvertently applying regularization pressure that the dataset cannot productively absorb. Unguided regularization on Stroke at $\lambda = 2$ inflicts a 10.47% accuracy loss with zero fairness benefit. Unguided EO regularization on Adult Income worsens DP disparity while partially improving EO. Unguided DP regularization on Diabetes produces spurious accuracy gains that mask model degradation.

In each case, the harm is not that regularization failed to improve fairness; it is that regularization was applied to a condition the dataset did not have. The adaptive weights, derived from BRD and PCS computed before training begins, govern whether regularization pressure is warranted, scaled, or suppressed. This is optimization governance: the diagnostic framework as a pre-training decision procedure that determines not how much fairness improvement to pursue, but whether pursuing it is structurally viable. This framing is consistent with risk-based formulations of fairness that characterize regularization effectiveness as a function of the underlying data distribution (Williamson and Menon 2019). Blind deployment of fairness interventions may itself constitute an ethical risk if predictive reliability is degraded without measurable equity benefit.

A further implication concerns hyperparameter non-transferability. $\lambda = 2$ produces a viable operating point on Heart Disease but catastrophic conflict on Adult Income, unnecessary degradation on Stroke, and metric artifacts on Diabetes. BRD and PCS explain why: they determine the effective gradient geometry, which differs fundamentally across datasets. The adaptive framework addresses this directly by making regularization strength a function of dataset structure rather than a fixed practitioner choice.

5.3 Implications for Fairness Evaluation in Clinical Machine Learning

The results from Diabetes 130-US and Stroke surface a practical challenge that extends beyond the fairness regularization setting: under class imbalance, standard fairness metrics can reflect properties of the correction procedure rather than genuine group disparities. On Diabetes, where male and female patients have near-identical 30-day readmission rates ($BRD = 0.001$), the unweighted baseline model exhibits near-zero fairness disparities not because it is fair, but because it predicts the majority class for nearly all patients and therefore treats both groups uniformly in the worst possible way.

Applying inverse-frequency class weighting, a necessary correction for the model to function clinically, introduces apparent DP and EO disparities of 0.032 and 0.025 respectively. These values do not reflect differential treatment of groups by the model; they reflect differential sensitivity of the prediction distribution to the weighting correction. A practitioner who observes these values and applies fairness regularization is optimizing against a measurement artifact.

This finding has implications for clinical deployment contexts beyond the specific regularization setting studied here. Fairness auditing of clinical models should account for the interaction between class imbalance correction procedures and fairness metric computation, particularly in high-imbalance settings such as readmission prediction, mortality prediction, and rare adverse event detection where positive class rates commonly fall below 15%. Class imbalance in clinical datasets is pervasive and well-documented, and its interaction with model evaluation metrics remains an active area of concern (Johnson and Khoshgoftaar 2019).

Standard accuracy is an unreliable primary metric in these settings, as demonstrated by the apparent accuracy improve-

Dataset	BRD	PCS	DP Regime	EO Regime	Adaptive	Outcome
Heart Disease	0.297	0.084	Viable	Partial	Full / Scaled	Viable / Limited
Adult Income	0.199	0.037	Conflict	Conflict	Scaled / Off	Conflict / Ineffective
German Credit	0.075	0.201	Boundary	Viable	Scaled / Full	Noise / EO Reduction
Diabetes 130-US	0.001	0.063	Low-Nec.	Partial	Off / Scaled	Artifact / Marginal
Stroke	0.004	0.028	Low-Nec.	Low-Nec.	Off / Off	Noise / Ineffective

Table 11: Diagnostic summary across all five datasets. DP and EO regimes are predicted from BRD and PCS before any model training. Adaptive column shows the adaptive weight behavior for DP / EO respectively (Full = $w = 1$, Scaled = $0 < w < 1$, Off = $w = 0$). Outcome column summarizes the observed experimental behavior for DP / EO. Low-Nec. = Low-Necessity.

ments under regularization on Diabetes that AUC correctly identifies as model degradation (Figure 3). We recommend that AUC be reported alongside, or in preference to, accuracy as the primary performance metric for imbalanced clinical prediction tasks, and that fairness metrics be interpreted in light of BRD to distinguish genuine group disparities from correction artifacts.

A second implication concerns the directionality of equal opportunity disparities. Both Heart Disease and Stroke exhibited baseline EO configurations in which minority-group TPR_{Female} exceeded majority-group TPR at baseline. Standard EO regularization, formulated to lift minority-group TPR toward majority-group TPR, may exhibit limited effectiveness when disparity directionality differs from the standard minority-underperformance setting.

PCS predicts EO gradient signal availability but not disparity directionality. Practitioners should examine baseline subgroup TPR direction before applying EO regularization; upward pressure on an already-higher minority TPR may have no effect or may reduce minority performance at high λ . Incorporating TPR directionality into the diagnostic is an important direction for future refinement, discussed further in the Limitations section.

6 Societal Implications and Ethical Considerations

The findings of this study carry implications that extend beyond the technical fairness regularization setting and bear directly on how algorithmic fairness tools are evaluated, deployed, and communicated in clinical contexts.

Fairness regularization is not a universal safeguard. A recurring assumption in responsible AI deployment is that applying fairness constraints during model training constitutes a meaningful intervention against discriminatory outcomes. The results of this study complicate that assumption. On three of five evaluated datasets, standard fairness regularization produced unnecessary accuracy degradation without meaningful fairness improvement. On one dataset, it actively worsened the fairness metric it was designed to reduce.

Deploying a fairness-regularized model is not equivalent to deploying a fair model, the structural conditions of the dataset determine whether the intervention has any positive effect. Clinical practitioners and institutional review processes that treat fairness regularization as a checkbox compliance step risk deploying models that are simultaneously less accurate and no more equitable than their unregularized counterparts.

Diagnostic-first deployment reduces avoidable harm.

The adaptive framework proposed in this work embodies a broader principle with direct ethical relevance: fairness interventions should be preceded by an assessment of whether the intervention is warranted. Applying regularization that degrades model accuracy without improving fairness does not constitute responsible deployment, it constitutes an avoidable harm to all patients whose outcomes depend on model predictions, with no compensating benefit to the groups the intervention was intended to protect.

The pre-training diagnostic proposed here requires no model training and adds negligible computational cost. Making such diagnostics a standard preliminary step in fairness-aware clinical ML development represents a concrete, actionable improvement to current practice.

Fairness interventions may affect underserved populations differently, and approaches that reduce disparity for one group may amplify it for unrepresented populations. This risk is compounded when interventions are deployed without community consultation, as affected groups are rarely involved in specifying fairness constraints.

We recommend that diagnostic frameworks such as the one proposed here be treated as inputs to a broader clinical governance process that includes domain expertise, community input, and ongoing monitoring of deployed model behavior across patient subpopulations. The ethical deployment of ML in healthcare requires ongoing attention to how algorithmic decisions interact with existing structural inequities in access to care (Chen, Joshi, and Ghassemi 2021; Obermeyer et al. 2019).

Pre-training diagnostics can inform the decision of whether to apply regularization, but they cannot substitute for the human judgment and institutional accountability that responsible clinical ML deployment requires.

7 Limitations

The proposed framework has two primary scope limitations: validation on five datasets and evaluation using logistic regression only. It should therefore be interpreted as an empirical diagnostic approach rather than a theoretically complete characterization, with several scope boundaries constraining generalizability.

Dataset scale and empirical thresholds. The diagnostic thresholds proposed in Equation 14 and Equation 15, $BRD \in \{0.05, 0.25\}$ and $PCS \in \{0.05, 0.15\}$, were derived empirically from five datasets. While the consistency of optimization behavior across these datasets pro-

vides initial support for the framework, the threshold values should be interpreted as empirically motivated approximations rather than universal constants.

With five datasets, overfitting of the diagnostic thresholds cannot be excluded, German Credit’s boundary behavior illustrates that datasets near thresholds may resist clean categorical assignment. Validation across substantially larger and more diverse datasets is required before the thresholds can be treated as prescriptive.

Model class and theoretical grounding. All experiments were conducted using logistic regression trained via full-batch gradient descent. This choice isolates the behavior of the fairness penalty formulations from confounding architectural effects, but limits the direct applicability of the findings to more expressive non-linear model classes such as neural networks and gradient boosting methods.

The gradient-conflict mechanism underlying BRD–DP may behave differently in deep non-linear representations, and the piecewise-linear adaptive weights were designed for interpretability rather than theoretical optimality, as demonstrated by the small accuracy degradation for adaptive DP on German Credit. The study does not provide formal convergence guarantees, the non-monotonic behavior observed on Heart Disease illustrates that fairness-aware optimization dynamics are not straightforwardly predictable from standard gradient descent theory. Extending to non-linear model classes with convergence guarantees remains an important direction for future work.

Fairness scope and sensitive attribute representation. The framework was evaluated exclusively with binary sensitive attributes (sex/gender), and all fairness metrics were computed for a single sensitive attribute at a time. Real-world clinical populations exhibit intersecting demographic characteristics: including race, age, socioeconomic status, and disability status, that may interact in ways not captured by single-attribute binary analysis. Intersectional demographic effects may therefore produce disparities that remain invisible under the current formulation.

The study focuses on DP and EO as representative fairness objectives. Equalized Odds, calibration-based, individual, and counterfactual fairness may exhibit substantially different optimization behavior. The diagnostic framework applies specifically to penalty-based in-processing methods targeting DP and EO; applicability to other paradigms should not be assumed without empirical validation.

EO directionality. The framework predicts EO gradient signal availability from PCS but does not model the directionality of baseline EO disparities. As demonstrated empirically in the Heart Disease 4.1 and Stroke 4.4 subsections above in which minority-group TPR exceeded majority-group TPR at baseline, a reversed disparity structure under which standard EO regularization may exhibit limited effectiveness.

PCS correctly predicted that sufficient gradient signal was available on these datasets, but it does not determine whether that signal can be productively deployed under the specific subgroup geometry present at baseline. Incorporating baseline TPR directionality alongside PCS represents one of the most actionable refinements to the current framework.

Associational and preprocessing constraints. The framework operates on associational fairness metrics, Demographic Parity and Equal Opportunity are defined in terms of statistical associations between predictions and sensitive attributes rather than causal relationships. BRD measures observed group-level outcome differences but does not distinguish between underlying causal differences and confounding effects; extending toward causal fairness analysis would require explicit causal graph specification beyond the scope of this work.

Additionally, BRD and PCS are computed from the pre-processed dataset, meaning their values depend on data preparation decisions including class weighting, imputation, and feature selection. Different preprocessing pipelines applied to the same raw data may yield different diagnostic values and consequently different adaptive weight assignments, a practical sensitivity that warrants systematic evaluation in future deployment studies.

8 Conclusion

Fairness-aware machine learning has made substantial progress in formalizing what it means for a model to treat demographic groups equitably. Less progress has been made on a prior question: whether a given fairness intervention is structurally viable for a given dataset before training begins. Current practice often assumes that gradient-based fairness regularization is broadly transferable across domains and prediction settings. The results of this work challenge that assumption directly.

Across five datasets spanning healthcare and non-healthcare domains, the same regularization procedure applied with the same optimization settings produced qualitatively different outcomes: a viable accuracy–fairness operating region on one dataset, sustained optimization conflict on another, and unnecessary accuracy degradation on others where baseline disparities were already limited. These differences were observed consistently despite using the same optimization procedure and hyperparameter configuration across datasets. The results instead aligned closely with two measurable structural properties of the data: Base Rate Disparity (BRD), which characterizes group-level outcome differences present in the training distribution, and Positive Class Sparsity (PCS), which characterizes the effective strength of EO gradient signal during optimization.

Pre-training predictions derived from these properties consistently aligned with observed optimization behavior across the evaluated datasets, supporting the central thesis of this work: fairness-aware optimization viability is governed by dataset structure.

This finding has a concrete design implication. The adaptive loss function proposed in this work translates the diagnostic properties directly into the training objective, modulating regularization strength according to the structural conditions associated with productive fairness optimization. The contribution of this adaptive formulation is not that it universally achieves better fairness outcomes than standard regularization. Rather, it provides a mechanism for governing fairness pressure according to dataset structure.

On datasets where regularization was warranted, the adaptive and standard formulations behaved similarly. On datasets where regularization was structurally unnecessary or ineffective, the adaptive framework reduced avoidable predictive degradation while preserving the meaningful fairness improvements observed under standard regularization. The distinction between fairness maximization and fairness governance captures the central practical contribution of this work.

BRD and PCS are computable directly from dataset statistics before training and provide a principled starting point for determining whether fairness-aware optimization is viable, ineffective, or unnecessary for a given dataset. Three concrete implications follow for practitioners deploying fairness-aware models in clinical settings.

Compute BRD and PCS before applying regularization. If $PCS < 0.05$, EO regularization will have insufficient gradient signal to exert meaningful influence regardless of λ . If $BRD < 0.05$ and baseline disparity is already small, DP regularization is unnecessary and will introduce noise. If BRD is moderate to high with sufficient PCS, proceed with regularization and tune λ on the dataset at hand.

Do not transfer λ across datasets. The same $\lambda = 2$ that produces a viable accuracy–fairness operating point on Heart Disease causes catastrophic conflict on Adult Income, unnecessary degradation on Stroke, and metric artifacts on Diabetes 130-US. BRD and PCS explain why: they determine the effective gradient geometry, which is a property of the data, not of the regularization method. Dataset-specific λ selection, informed by the diagnostic profile, is the correct approach.

Monitor AUC alongside accuracy during fairness-aware training. Accuracy can degrade while AUC remains stable, indicating the model retains discriminative capacity even when binary decisions become less reliable, as observed consistently across all five datasets in this study. In imbalanced clinical settings, accuracy can also appear to improve under regularization while AUC declines, as demonstrated on Diabetes 130-US, masking genuine model degradation. AUC is the more reliable indicator of whether fairness-aware training is helping or harming the model’s clinical utility.

The framework presented here remains empirical rather than theoretical, and the threshold values should be interpreted as approximations rather than universal constants. Future work should evaluate the framework across larger datasets and non-linear model classes.

What this study establishes is empirical evidence that fairness-aware optimization behavior may depend systematically on measurable structural properties of the dataset. Making this dependence explicit enables principled pre-training fairness analysis.

References

Agarwal, A.; Beygelzimer, A.; Dudik, M.; Langford, J.; and Wallach, H. 2018. A Reductions Approach to Fair Classification. In *Proceedings of the 35th International Conference on Machine Learning*, 60–69.

Barocas, S.; Hardt, M.; and Narayanan, A. 2019. *Fair-*

ness and Machine Learning: Limitations and Opportunities. fairmlbook.org.

Berk, R.; Heidari, H.; Jabbari, S.; Joseph, M.; Kearns, M.; Morgenstern, J.; Neel, S.; and Roth, A. 2017. A Convex Framework for Fair Regression. *arXiv preprint arXiv:1706.02409*.

Chen, I. Y.; Joshi, S.; and Ghassemi, M. 2021. Ethical Machine Learning in Healthcare. *Annual Review of Biomedical Data Science*, 4: 123–144.

Chouldechova, A. 2017. Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments. *Big Data*, 5(2): 153–163.

Corbett-Davies, S.; and Goel, S. 2018. The Measure and Mismeasure of Fairness: A Critical Review of Fair Machine Learning. *arXiv preprint arXiv:1808.00023*.

Cotter, A.; Jiang, H.; and Sridharan, K. 2019. Optimization with Non-Differentiable Constraints with Applications to Fairness, Recall, Churn, and Other Goals. In *Journal of Machine Learning Research*, volume 20, 1–59.

Donini, M.; Oneto, L.; Ben-David, S.; Shawe-Taylor, J.; and Pontil, M. 2018. Empirical Risk Minimization Under Fairness Constraints. In *Advances in Neural Information Processing Systems*, volume 31.

Dwork, C.; Hardt, M.; Pitassi, T.; Reingold, O.; and Zemel, R. 2012. Fairness Through Awareness. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*, 214–226.

Fawcett, T. 2006. An Introduction to ROC Analysis. *Pattern Recognition Letters*, 27(8): 861–874.

Finocchiaro, J. 2023. Using Property Elicitation to Understand the Impacts of Fairness Regularizers. *arXiv preprint arXiv:2309.11343*.

Hardt, M.; Price, E.; and Srebro, N. 2016. Equality of Opportunity in Supervised Learning. In *Advances in Neural Information Processing Systems*, volume 29.

He, H.; and Garcia, E. A. 2009. Learning from Imbalanced Data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9): 1263–1284.

Johnson, J. M.; and Khoshgoftaar, T. M. 2019. Survey on Deep Learning with Class Imbalance. *Journal of Big Data*, 6(1): 1–54.

Kamishima, T.; Akaho, S.; Asoh, H.; and Sakuma, J. 2012. Fairness-aware Learning through Regularization Approach. In *Proceedings of the IEEE 11th International Conference on Data Mining Workshops*, 643–650.

Kleinberg, J.; Mullainathan, S.; and Raghavan, M. 2017. Inherent Trade-Offs in the Fair Determination of Risk Scores. In *Proceedings of the 8th Innovations in Theoretical Computer Science Conference*.

Martinez, N.; Bertran, M.; and Sapiro, G. 2020. Minimax Pareto Fairness: A Multi Objective Perspective. In *Proceedings of the 37th International Conference on Machine Learning*, 6755–6764.

Mehrabi, N.; Morstatter, F.; Saxena, N.; Lerman, K.; and Galstyan, A. 2021. A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys*, 54(6): 1–35.

- Obermeyer, Z.; Powers, B.; Vogeli, C.; and Mullainathan, S. 2019. Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations. *Science*, 366(6464): 447–453.
- Pföhl, S. R.; Foryciarz, A.; and Shah, N. H. 2021. An Empirical Characterization of Fair Machine Learning for Clinical Risk Prediction. *Journal of Biomedical Informatics*, 113: 103621.
- Rajkomar, A.; Oren, E.; Chen, K.; et al. 2018. Scalable and Accurate Deep Learning with Electronic Health Records. *npj Digital Medicine*, 1(1): 18.
- Sener, O.; and Koltun, V. 2018. Multi-Task Learning as Multi-Objective Optimization. In *Advances in Neural Information Processing Systems*, volume 31.
- Williamson, R. C.; and Menon, A. K. 2019. Fairness Risk Measures. In *Proceedings of the 36th International Conference on Machine Learning*, 6786–6797.
- Zafar, M. B.; Valera, I.; Gomez-Rodriguez, M.; and Gum-madi, K. P. 2017. Fairness Constraints: Mechanisms for Fair Classification. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, 962–970.